

1.1.2 Option B – Glaciated Landscapes

1. How can glaciated landscapes be viewed as systems?	
Key Ideas	Content
1.a. Glaciated landscapes can be viewed as systems.	- A conceptual overview of: - the components of glaciated landscape systems, including inputs, processes and outputs - the flows of energy and material through glaciated systems - glacier mass balance.
1.b. Glaciated landscapes are influenced by a range of physical factors.	- Potential influences on glaciated landscape systems of: - climate, including precipitation totals and patterns - geology, including lithology and structure - latitude and altitude - relief and aspect on microclimate and glacier movement.
1.c. There are different types of glacier and glacier movement.	- The characteristics of different types of glacier and their movement, including: - the formation of glacier ice - valley glaciers and ice sheets - warm-based and cold-based glaciers - basal sliding and internal deformation.

2. How are glacial landforms developed?	
Key Ideas	Content
2.a. Glacial landforms develop due to a variety of interconnected climatic and geomorphic processes.	- The influence of flows of energy and materials on geomorphic processes, including weathering, mass movement, glacial erosion, nivation, transportation and deposition. - The formation of distinctive landforms, predominantly influenced by erosion, including corries, arêtes, pyramidal peaks, troughs, roche moutonnée and striations. - The formation of distinctive landforms, predominantly influenced by deposition, including terminal, lateral and recessional moraines, erratics, drumlins and till sheets.
2.b. Glacial landforms are inter-related and together make up characteristic landscapes.	Case studies of one landscape associated with the action of valley glaciers and one associated with the action of ice sheets to illustrate: - the physical factors which influence the formation of landforms within the landscape system - the inter-relationship of a range of landforms within the characteristic landscape system - how and why the landscape system changes over time from millennia to seconds, such as rock fall in seconds, seasonal changes in deposition rates and erosion of basins over millennia. At least one of the case studies must be from beyond the UK.

3. How do glacial landforms evolve over time as climate changes?

Key Ideas	Content
3.a. Glacio-fluvial landforms exist as a result of climate change at the end of glacial periods.	• How landforms in glaciated landscapes are influenced in post-glacial periods, including: ◦ climate changes that occurred during a post-glacial period and the effect on resultant geomorphic processes ◦ the influence of these processes in forming landforms, including kames, eskers and outwash plains ◦ the modification of these landforms by processes associated with present and future climate changes.
3.b. Periglacial landforms exist as a result of climate change before and/or after glacial periods.	• How landforms in periglacial landscapes are influenced by climate change, including: ◦ climate changes that occurred during a previous time period and the effect on resultant geomorphic processes ◦ the influence of these processes in forming landforms, including patterned ground and pingos ◦ the modification of these landforms by processes associated with present and future climate changes.

4. How does human activity cause change within glaciated and periglacial landscape systems?

Key Ideas	Content
4.a. Human activity causes change within periglacial landscape systems.	• Case study of one periglacial landscape that is being used by people, to illustrate: ◦ the human activity taking place and the reasons for it taking place, such as resource extraction ◦ the impacts on processes and flows of material, processes and/or energy through the periglacial system, such as increased heat produced by buildings ◦ the effect of these impacts in changing periglacial landforms, such as thawing of permafrost ◦ the consequence of these changes on the landscape, such as development of thermokarst.
4.b. Human activity causes change within glaciated landscape systems.	• Case study of one glaciated landscape that is being used by people, to illustrate: ◦ the human activity taking place and the reasons for it taking place, such as dam construction ◦ the impacts on processes and flows of material, processes and/or energy through the glacial system, such as trapping of sediment ◦ the effect of these impacts in changing glacial landforms, such as increased channel scour below dams ◦ the consequence of these changes on the landscape, such as changes to valley floor.

5. Topic-specific skills:

- observation skills
- measurement and geo-spatial mapping skills
- data manipulation and statistical skills applied to field measurements
- sediment budget calculations
- mass balance calculations.